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Mapping Anti-HLA Class I Cross-Reactivity for Transplantation Using Interpretable Embedding and Clustering of SAB MFI

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Abstract

Background: Mapping anti-HLA class I cross-reactivity from single-antigen bead (SAB) mean fluorescence intensity (MFI) data supports donor selection. However, interpretation is complicated by analytical choices and assay variability. **Methods:** A total of 4327 SAB assays were analyzed (antigen × test matrix) using an interpretable, distance-based workflow. Antigen profiles were z-scored across tests. Multidimensional scaling (MDS) was used for visualization and hierarchical clustering analysis (HCA) for grouping, and complemented these with a common PCA space for model selection (K-Means via Silhouette; Gaussian Mixture Models via BIC), agglomerative (Ward and average-link on 1–correlation), spectral clustering on correlation-derived affinities, and a graph approach (k-NN ∪ minimum-spanning-tree with modularity-based communities). Non-linear embeddings (t-SNE/UMAP) and density-based HDBSCAN were used only for visual analytics, not for primary inference. **Results:** The pipeline revealed coherent reactivity neighborhoods that partially overlapped known cross-reactive antigen groups (CREGs) and eplet-based expectations while also highlighting less-documented relationships. Robustness was confirmed through bootstrap resampling, graph modularity, and consensus clustering across methods. **Conclusions:** This unified, auditable workflow converts descriptive maps into method-robust summaries of antibody reactivity and cross-reactivity. While exploratory and performed on a single dataset without linked outcomes, the approach provides a reproducible structure for comparing cohorts and prioritizing hypotheses that could be prospectively validated for clinical decision support in transplantation.

Keywords: HLA Class I; single-antigen bead; mean fluorescence intensity; multidimensional scaling; hierarchical clustering; computational immunology; interpretable visualization; cross-reactivity mapping

Key Contribution: This study summarizes anti-HLA Class I reactivity into groups at an interpretable cut-level using hierarchical clustering and provides distance-preserving, interpretable embeddings via multidimensional scaling. The combined, explainable visualizations contextualize known cross-reactive relationships and highlight less-described proximities that warrant targeted confirmation.



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1. Introduction

Transplant outcomes are strongly modulated by alloimmune recognition of donor HLA, and anti-HLA class I antibodies remain central to risk assessment [1–6]. Single-antigen bead (SAB) assays summarize reactivity as mean fluorescence intensity (MFI), but interpretation depends on analytical choices and assay variability [7–10]. This study asks whether a transparent, distance-based workflow can organize SAB MFI into reproducible neighborhoods that align with known biology (CREGs/eplets) while revealing less-documented proximities. We use multidimensional scaling (MDS) for visualization and hierarchical clustering (HCA) for grouping, and we corroborate structure with complementary unsupervised methods and stability analyses. The goal is not a new “black-box” model, but an auditable map clinicians can interpret and that can be compared across cohorts.

The high degree of polymorphism in HLA genes results in a diverse array of HLA Class I molecules, which is beneficial for a broad range of immune responses against various pathogens, but also poses significant challenges in transplantation. This genetic diversity can manifest as alloreactivity (immune reaction against donor HLA molecules) and cross-reactivity, the latter of which is particularly important in transplantation immunology. Cross-reactivity refers to the immune system’s response not only to specific donor HLA epitopes but also to previously encountered similar epitopes. This phenomenon can be analyzed through patterns of antibody reactivity, which are often grouped into Cross-Reactive Groups (CREGs). These groups represent sets of HLA antigens that share structural similarities, leading to cross-reactive antibody responses. By understanding these patterns and identifying CREGs through MFI data, clinicians can better predict and manage potential cross-reactive responses during organ transplantation [11–15].

Moreover, the specific epitopes on HLA molecules, which are recognized by the immune system, play crucial roles in both alloreactivity and cross-reactivity. Epitopes are the parts of the antigen that antibodies or T-cell receptors bind to. In the context of organ transplantation, the precise identification of these epitopes and their roles in immune responses is vital. Advancements in epitope analysis and matching, such as sequence-based typing, have enhanced the precision of HLA compatibility assessments. This level of detail is essential for improving transplant outcomes and managing post-transplant immunosuppression more effectively [16].

The evolving field of transplantation immunology has seen significant advancements in HLA analysis techniques. Modern approaches have advanced beyond simple HLA typing to include more intricate methods that consider the entire range of possible immune responses [17]. This includes the assessment of donor-specific antibodies (DSAs), which are critical for predicting the risk of antibody-mediated rejection. DSAs are antibodies that specifically target the HLA molecules of the transplant donor. The presence and specificity of these antibodies, as revealed by MFI values, are strong predictors of transplant success or failure [10,18].

The use of MFI for DSA assessment has revolutionized the pre-transplant evaluation process, allowing for a more tailored approach to donor selection and immunosuppressive therapy. By quantifying antibody levels against specific HLA antigens, clinicians can identify patients at high risk of rejection and adjust treatment protocols accordingly. This personalized approach to transplantation, guided by MFI data, considerably improves the likelihood of successful transplantation and long-term graft survival [7,19–22].

Furthermore, MFI values play a crucial role in post-transplant monitoring. Regular assessment of antibody levels via MFI can provide an early warning of potential rejection episodes, thereby facilitating timely intervention. This monitoring is particularly critical in the early post-transplant period when the risk of acute rejection is highest [23,24].

Prior analyses of SAB matrices have used hierarchical or heatmap-based grouping to visualize co-reactivity and infer cross-reactive relationships, and epitope-centric frameworks (e.g., eplet analysis) increasingly ground these observations mechanistically [15,25,26]. Our contribution is not a novel learner per se, but a unified, auditable workflow that (i) uses distance-preserving embedding to expose neighborhoods, (ii) selects a partition scale with complementary evidence, (iii) stress-tests structure across clustering families and a graph-modularity view, (iv) quantifies stability, and (v) reports a cross-method consensus co-association matrix. This triangulation turns a descriptive map into a method-robust summary suitable for comparison across cohorts.

In conclusion, the interplay between alloreactivity, cross-reactivity, epitope recognition, and the identification of CREG patterns in HLA Class I molecules constitutes a complex and crucial aspect of organ transplantation. The precise measurement and interpretation of MFI values in this context have become indispensable in modern transplantation medicine, playing a pivotal role in ensuring successful transplantation procedures, enhancing patient outcomes, and minimizing the risk of organ rejection. Combining SAB assays with explainable embedding and clustering enables transparent, reproducible summaries of reactivity patterns suited to hypothesis generation in transplantation immunology [27].

The aim of this study is to apply advanced analytical techniques, including multidimensional scaling (MDS) and hierarchical cluster analysis (HCA), to interpret anti-HLA antibody reactivity patterns. By leveraging these methods on MFI data, this study seeks to identify CREGs and provide a comprehensive understanding of antibody reactivity profiles, ultimately supporting improved strategies for donor-recipient matching and post-transplant monitoring. This study demonstrates how embedding and clustering techniques, widely used in other high-dimensional biomedical domains, can be adapted for immunogenetics to produce transparent, reproducible visualizations that support hypothesis generation and facilitate integration into data-driven clinical research.

2. Materials and Methods

2.1. Data Source and Preparation

This study focused on MFI data sourced from Class I SAB assays, which are essential for identifying and performing a semi-quantitative analysis of anti-HLA antibodies. The initial dataset comprised a comprehensive matrix representing MFI values aligned with a range of HLA antigens per sample. A total of 4327 validated SAB assays were included, with each performed using the LABScreen Single Antigen Class I Assay (LS1A04 lot 14, One Lambda, West Hills, CA, USA) according to the manufacturer's recommendations. To avoid the prozone phenomenon, ethylenediaminetetraacetic acid (EDTA) was added to a final concentration of 0.005 M. Negative control beads lacking HLA antigens were used to monitor non-specific signal. This study did not apply a fixed positivity cut-off; analyses used raw MFI and interpreted reactive neighborhoods relative to the negative-bead baseline.

Analyses used raw MFI interpreted relative to the negative-bead baseline to avoid thresholding artifacts. Because SAB MFI is semi-quantitative and sensitive to lot/instrument drift and prozone, all assays followed a consistent protocol (same panel lot, instrument family, EDTA 0.005 M). Standardization (z-scoring by antigen across tests) and correlation-based sensitivity analyses were used to attenuate scale and technical variability. While minor operational variability is unavoidable in real-world clinical testing, our standardization and correlation-based sensitivity analyses are designed to attenuate its impact on inferred structure.

2.2. Constructing the Dissimilarity Matrix

The initial step in this analytical process was to construct a dissimilarity matrix using the Euclidean distance metric, which quantifies differences in antigen reactivity profiles. This matrix was crucial because it served as the primary input for the subsequent MDS analysis, which aimed to simplify the complex immunological data into a more interpretable, reduced-dimensional format. Although Euclidean distance was used for the primary MDS/HCA, correlation-based dissimilarity ($1-r$) as a sensitivity analysis (Sections 2.8.1 and 3.5.4) was also evaluated.

2.3. Execution of Multidimensional Scaling

This MDS approach aims to transpose intricate high-dimensional MFI data into a simplified two-dimensional space. For this purpose, the scikit-learn library's MDS algorithm (version 1.0.2) was employed. Computational limitations required optimization of the algorithm's parameters: the analysis set the number of dimensions to two ($n_components = 2$), limited the number of algorithm initializations to one ($n_init = 1$), and capped the maximum number of iterations at 100 ($max_iter = 100$). To ensure consistency in the results, a predetermined random state ($random_state = 42$) was used. MDS places antigens so that inter-point distances approximate the original dissimilarities, providing an interpretable proximity map. Reproducibility was ensured by fixing random seeds and detailing code implementations in a manner compatible with integration into broader clinical decision-support research pipelines.

2.4. Stress and Goodness-of-Fit Metrics

To evaluate the MDS model's accuracy, two key metrics were employed: stress and R-square. Stress provided a quantification of the discrepancies between the high-dimensional dissimilarities and the MDS plot distances. In addition to Kruskal's stress, R^2 was computed as the squared Pearson correlation between original dissimilarities and embedded inter-point distances, providing an intuitive goodness-of-fit summary. Here, "high-dimensional dissimilarities" refers to distances computed directly on the original antigen reactivity profiles, prior to embedding. The purpose of MDS is to preserve these dissimilarities as faithfully as possible in the reduced 2D or 3D representation.

2.5. MFI Correlation Analysis

A correlation analysis was conducted between different anti-HLA antibody MFI values to identify cross-reactivity patterns and similarities in the reactivity profiles. Using Pearson's correlation coefficient, the analysis measured the linear relationships between variables, with coefficients ranging from -1 (perfect negative linear relationship) to $+1$ (perfect positive linear relationship).

2.6. Correlation Data Interpretation

The resulting correlation matrix was essential in deciphering antigenic similarities among the antibodies. Strong positive correlations hinted at the likelihood of certain antibodies binding to similar epitopes. Conversely, strong negative correlations indicate potential competitive binding dynamics. This matrix also laid the groundwork for subsequent analyses, like hierarchical clustering, to classify antibodies based on their reactivity patterns.

2.7. Statistical Software

For this study, various libraries were utilized, including Python's Pandas library (version 1.2.3) for data manipulation and correlation analysis, Seaborn (version 0.13.2) and

Matplotlib (version 3.9.2) for generating comprehensive heatmaps, and Scikit-learn for executing MDS, which together enabled a detailed exploration of anti-HLA antibody patterns.

2.8. Extended Analytic Pipeline (Robustness and Complementary ML)

To assess whether the inferred reactivity structure is robust to modeling choices and to complement MDS/HCA with contemporary unsupervised methods, the following pipeline on the antigen-by-test matrix was implemented. Unless stated otherwise, all analyses were conducted in Python using standard scientific libraries.

2.8.1. Preprocessing and Feature Scaling

Raw SAB MFI values were organized as an antigen (features) $\times$ test (samples) matrix. Each antigen's profile was z-scored across tests (mean 0, SD 1) to mitigate scale effects and attenuate assay-to-assay variability (e.g., lot or instrument drift). Where noted, correlation-based distances ($1 - \text{Pearson's } r$) were used as a sensitivity analysis because they down-weight absolute intensity shifts.

2.8.2. Dimensionality Reduction (PCA)

Principal component analysis (PCA) was fit on the standardized matrix. This study retained up to 20 components (or at most $n_{\text{tests}} - 1$) and used this compact space as the common representation for all complementary models. Cumulative variance explained was recorded to document information retention.

2.8.3. Partitioning/Model-Based Clustering and Model Selection

Two clustering families on the PCA space for $k = 2-12$ were screened:

- i. K-Means (partitioning);
- ii. Gaussian Mixture Models ((GMM) full-covariance, model-based).

For K-Means, candidate k values were scored by the mean Silhouette on PCA features; for GMMs, model evidence was assessed by the Bayesian Information Criterion (BIC). This study first selected a single summary k where both criteria agreed; when they differed, the analysis favored the k at which Silhouette improvements plateaued while BIC was near its minimum (precluding over-fragmentation). The “winner” algorithm at the adopted k was then chosen by the higher Silhouette computed on the common feature space. All fits used fixed random seeds and multiple initializations for reproducibility.

Rationale for k and clinical interpretability. The working number of clusters was not fixed a priori. This study scanned $k = 2-12$ and combined complementary criteria: the mean Silhouette for K-Means and the BIC for GMM. This study adopted the smallest k at the Silhouette “elbow” that was also near the BIC minimum, balancing granularity with parsimony. At this scale, clusters were sufficiently coarse to remain clinically interpretable while capturing consistent mesoscale structure seen across methods (Sections 3.5.2–3.5.5). Where relevant, concordance with canonical CREGs and eplet sharing was described in the Results (Sections 3.1–3.4).

2.8.4. Stability Analysis (Bootstrap AMI)

Robustness to sampling variability was quantified via five bootstrap resamples of tests (80% with replacement). For each resample, the selected model was fit twice with different seeds, and stability was summarized by the adjusted mutual information (AMI) between the two labelings.

2.8.5. Alternative Clustering Families

To evaluate sensitivity to distance/modeling choices, the analysis additionally fit:

- i. Agglomerative clustering with Ward linkage on Euclidean PCA space;
- ii. Agglomerative clustering with average linkage on a precomputed correlation distance ($1-r$);
- iii. Spectral clustering on a correlation-derived affinity matrix.

For comparability across algorithms, cluster quality indices (Silhouette, Calinski–Harabasz, Davies–Bouldin) were computed on the same feature space.

2.8.6. Graph Construction and Community Detection

An antigen–antigen similarity network by uniting a positive-correlation k-nearest-neighbor graph with a minimum-spanning-tree backbone to ensure connectivity was constructed, and the resulting undirected graph retained non-negative edge weights. Communities were identified by greedy modularity maximization (primary analysis). Where available, Louvain and Leiden algorithms were applied as sensitivity checks on the same graph. Modularity (Q) was used to summarize mesoscale organization.

2.8.7. Non-Linear Embeddings and Density-Based Grouping (Visual Analytics)

For visualization and hypothesis generation only, t-distributed stochastic neighbor embedding (t-SNE) and, when available, uniform manifold approximation and projection (UMAP) on the PCA features were computed. To highlight dense neighborhoods, hierarchical density-based spatial clustering of applications with noise (HDBSCAN) was optionally applied on the embedding; “noise” labels were treated as unassigned. These steps were not used for primary quantitative claims.

2.8.8. Consensus Clustering Across Methods

Agreement across algorithms was summarized by a co-association matrix, defined as the fraction of methods that co-assign each antigen pair to the same cluster (considering the winner, spectral, Ward, average-correlation, and, when present, HDBSCAN labelings). Hierarchical clustering on $1 - \text{co-association}$ yielded a consensus partition used to corroborate modules observed in the winner model and graph communities.

2.8.9. Software and Reproducibility

Analyses used NumPy, pandas, scikit-learn, SciPy, and networkx; optional components included umap-learn, hdbscan, python-louvain, and igraph/leidenalg. Seeds were fixed where applicable, and multiple initializations were used. On Windows, thread management safeguards were enabled to avoid BLAS/thread-pool conflicts. All hyperparameters not explicitly stated followed library defaults.

2.8.10. Notes on Batch Effects and Scaling Choices

SAB MFI is semi-quantitative and sensitive to reagent lot, instrument settings, and matrix effects. To mitigate technical scale differences, each antigen profile was z-scored across tests before modeling, and correlation-based distances/affinities ($1-r$) were used in sensitivity analyses because they de-emphasize absolute intensity shifts. This study verified that principal neighborhoods persisted under Euclidean- and correlation-based formulations and across clustering families (Sections 3.5.2–3.5.4), which reduces the likelihood that structure reflects batch artifacts alone.

3. Results

This study focused on exploring the patterns of anti-HLA antibody reactivity using MFI data derived from SAB assays. The goal of this study was to understand these antibodies, with a focus on identifying potential cross-reactivity and interactions that might be implicated in humoral responses. The results encompass several key analyses, including HCA, heatmap visualization, 3D and 2D MDS, and correlation analysis. The combination of distance-preserving embeddings and hierarchical organization is intended to yield explainable outputs, with a visual structure that can be interpreted by both computational researchers and clinicians.

3.1. Hierarchical Clustering Analysis

HCA was applied to antigen profiles (MFI), Figure 1, producing a dendrogram in which shorter branch lengths reflect greater similarity. At a clinically interpretable cut-level, HCA grouped antigens with similar reactivity profiles; the number of groups follows the chosen cut and was used to describe the structure.

At more stringent distance thresholds, clusters became overly granular. At an interpretable cut-level, HCA grouped HLA antigens with similar reactivity profiles; the number of groups depends on the chosen cut-level and is used descriptively to summarize broader structure. These clusters help illustrate cross-reactivity patterns among anti-HLA antibodies and provide insight into related reactivity profiles. The resulting dendrogram visualizes these clusters, where the branch lengths reflect the degree of similarity between them. Shorter branches indicate a close relationship, pointing to shared epitopes or similar immune responses.

Adjusting the distance threshold can create more comprehensive clusters, highlighting commonalities among the assays. The HCA dendrogram allows for the observation of cluster formation and relationships among different anti-HLA assays. Clusters formed at shorter distances indicate assays with closely related MFI profiles. For instance, anti-HLA A*02:01, anti-HLA A*02:03, and anti-HLA A*02:06 assays appear to form tight clusters, suggesting a close relationship. This proximity could indicate shared reactivity patterns or similar immune responses. However, longer distances between antibodies suggest higher dissimilarity levels. An example is the distance between anti-HLA A*02:01 and anti-HLA A*68:01, which, despite being in the same cluster, cluster 1, had longer branch lengths, indicating more significant differences in MFI and similar types of response.

Interestingly, the formation of clusters 3 and 4, which are mainly populated by antibodies directed to the HLA C locus, highlights the proximity within this antibody profile and their dissimilarity from those arising from other HLA class I loci. Some cross-reactive groups identified in this study, such as those consistent with C1 and C2 supertypes, align with well-established classifications within HLA immunology. These findings underscore our methodology's potential for validating known patterns, while also highlighting less characterized interactions. Examining the patterns revealed by HCA can lead to the identification or clarification of which HLA antigens are likely to elicit cross-reactive antibody responses, thereby aiding in critical decisions regarding the transplantation process.

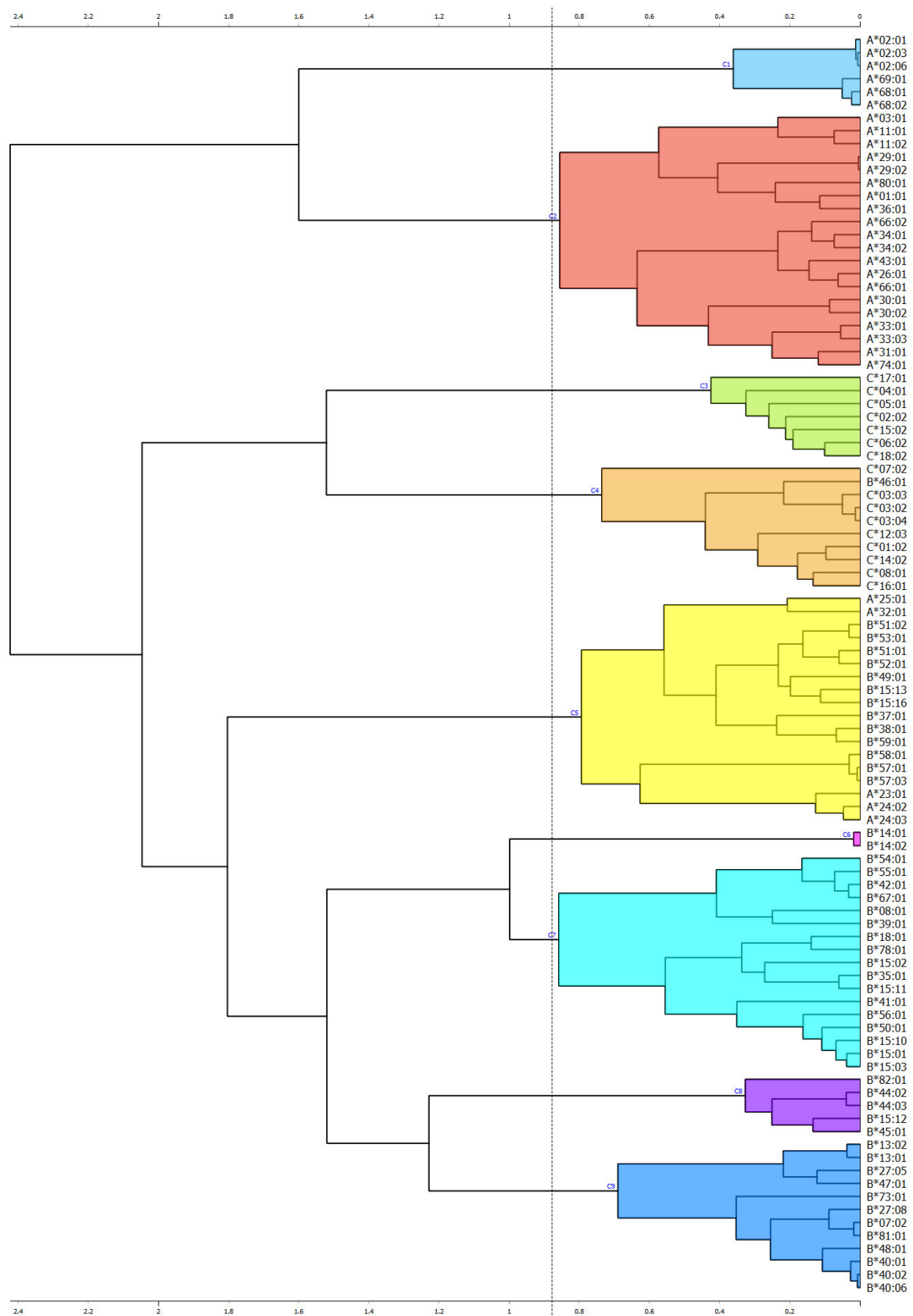


Figure 1. Hierarchical clustering of anti-HLA Class I reactivity (SAB MFI). Dendrogram of antigens clustered by similarity in their MFI profiles, cut at an interpretable level to define groups with shared reactivity patterns. Shorter branch lengths indicate closer similarity. The resulting clusters summarize antigen proximity, highlight putative cross-reactive neighborhoods, and provide an interpretable map to support expert review and guide follow-up analyses.

Furthermore, HCA allows for deeper analysis to reveal the nature of CREGs within the dataset, aiding in the identification of common reactivity profiles among various anti-HLA antibodies. A heatmap was generated to visualize the reactivity patterns across all anti-HLA antibodies. This heatmap provides a comprehensive view of the MFI data, where each column and row represents the interaction among specific antibodies. Although both the heatmap and HCA analyses are based on the same dissimilarity matrix, they serve complementary purposes. The HCA provides a structured view of specificities through automatic ordering, while the heatmap offers a color-coded visualization of reactivity intensities across all pairs, which is helpful for a quick overview of broader relationships. Matched clusters to historical CREGs and, when applicable, to eplet motifs (Supplementary Table S1), identifying A2/A68/A69 as a canonical 2C block and several Bw4-enriched B-locus modules.

Identifying these groups can help clinicians predict immune responses to transplantation and guide decisions about donor selection and post-transplant management. For example, cluster 2 reveals CREG 1A, which includes anti-HLA A*01, anti-HLA A*03, anti-HLA A*11, anti-HLA A*29, anti-HLA A*30, anti-HLA A*31, anti-HLA A*36, and anti-HLA A*80. Another example is cluster C5, which shows a response directed against eplet 80I, including A*23, A*24, A*25, A*32, B*15:13, B*15:16, B*15:17, B*15:24, B*38, B*49, B*51, B*52, B*53, B*57, B*58, and B*59.

3.2. Heatmap Analysis

This study visualized pairwise reactivity as a heatmap to complement the HCA ordering: rows/columns are antigens and color encodes MFI, where each column and row represents the interaction among specific antibodies, with the color intensity in the heatmap reflecting the MFI level, with red and blue shades indicating stronger and weaker reactivity, respectively.

Overall reactivity is shown in Figure 2.

The heatmap shows stronger reactivity among antibodies directed against antigens belonging to the same locus. This may occur due to several factors, including conserved epitopes across alleles in the same locus (e.g., conserved regions, CREGs); shared immunogenicity (e.g., immunodominance, where some alleles within the same locus present epitopes that are particularly immunodominant, leading to the production of a larger quantity of antibodies); or similarity in peptide binding, as alleles in the same locus often have similar peptide-binding preferences, which can elicit similar antibody responses. Additionally, technical and assay-specific factors, such as SAB assay sensitivity and antigen representation on the beads, may also contribute to the observed reactivity.

Figure 3 illustrates this pattern: panels 3a–3c highlight within-locus reactivity (HLA-A, HLA-B, HLA-C), whereas panels 3d–3f show weaker cross-locus relationships (A vs. B, A vs. C, B vs. C). For instance, similarity or clustering can be observed in Figure 3a for HLA-A*01:01 and HLA-A*36:01; HLA-A*03:01, HLA-A*11:01, and HLA-A*11:02, or even a strong correlation between HLA-A*02:01, HLA-A*02:03, and HLA-A*02:06 and a weaker but still important correlation with the previous ones and HLA-A*68:01, HLA-A*68:02, and HLA-A*69:01. Similarly, in locus C, several strong correlations could be observed, namely HLA-C*03:02, HLA-C*03:03, and HLA-C*03:04; and weaker correlations with the previous ones, namely HLA-C*08:01, HLA-C*12:03, and HLA-C*14:02. When comparing different loci, the only relationship that can be observed is between HLA-A and HLA-C loci (Figure 3e).

The addition of a third dimension to the MDS plot can enrich the analysis by allowing deeper inspection of relationships among anti-HLA antibodies. The points clustered closely together suggest a higher similarity in reactivity, indicating potential cross-reactivity.

Conversely, points that are positioned further apart denote greater dissimilarity, which suggests distinct reactivity patterns. This 3D perspective is invaluable for identifying clusters of antigens with similar reactivity profiles, thereby helping clinicians visualize broader patterns of cross-reactivity. These insights are crucial for refining donor–recipient matching and minimizing the risk of transplant rejection. The three-dimensional spatial arrangement provides a nuanced understanding of the interactions between different antigens, which is vital for the complex decision-making required in transplant medicine.

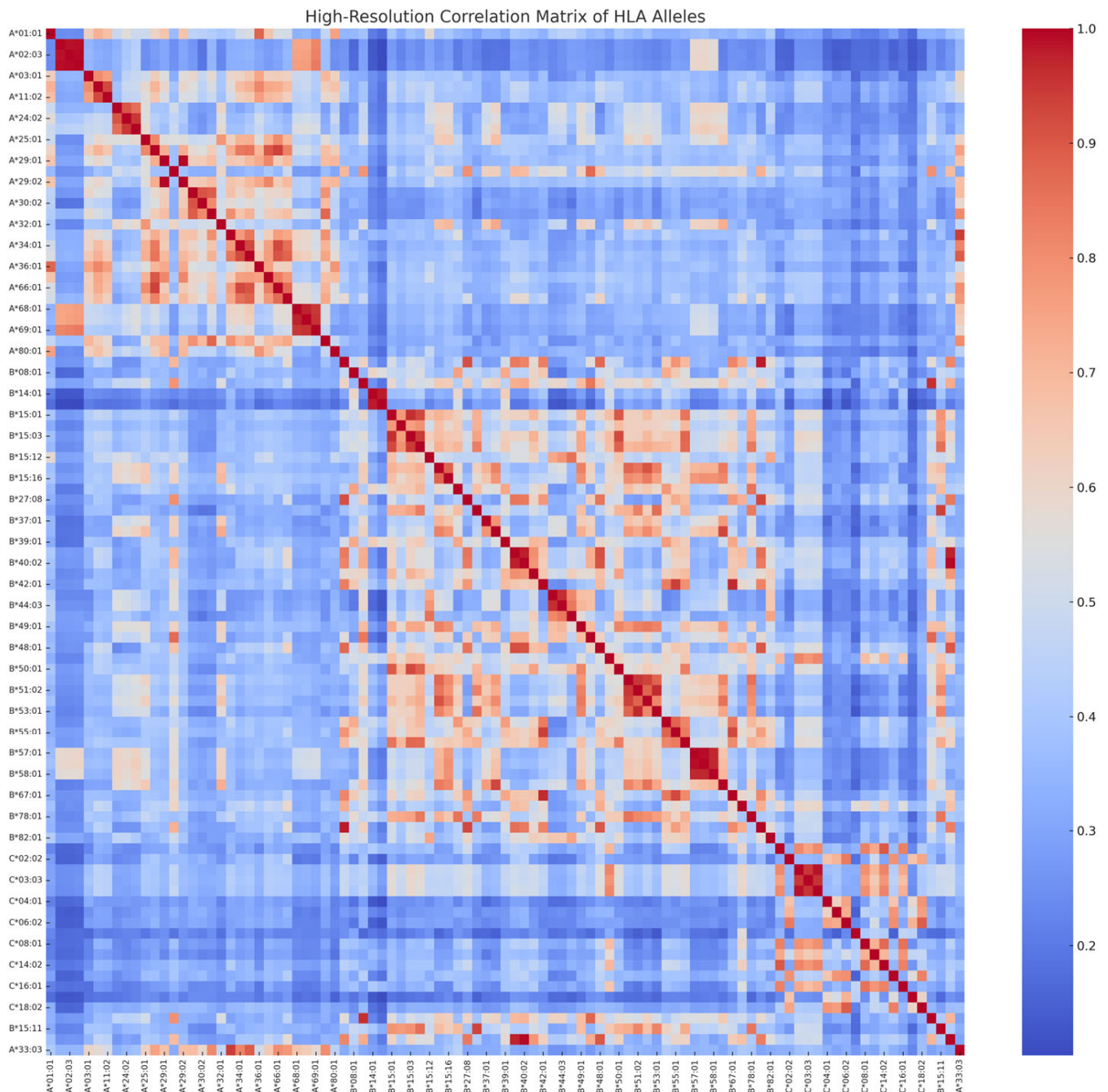


Figure 2. Heatmap of all anti-HLA class I assays (columns) against antigens (rows). Color encodes MFI (higher = stronger reactivity). The map highlights within-locus similarity and guides subsequent clustering and correlation analyses.

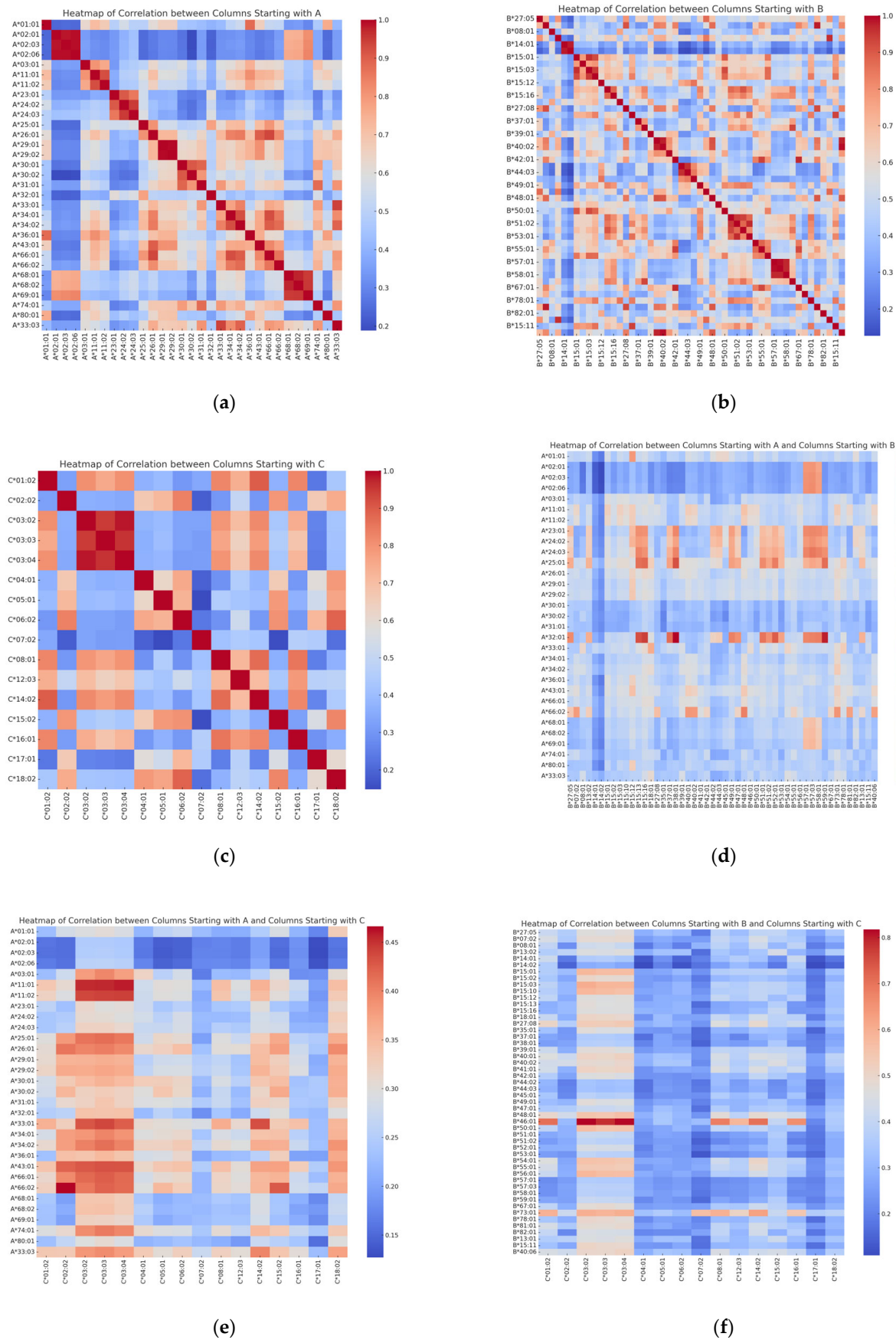


Figure 3. Locus-stratified heatmaps. (a–c) Within-locus reactivity for HLA-A, -B, -C; (d–f) cross-locus comparisons (A vs. B, A vs. C, B vs. C). Strongest structure appears within loci, with limited cross-locus signal.

3.3. Multidimensional Scaling

To further analyze the relationships among anti-HLA antibodies, they were evaluated employing both 2D and 3D MDS. This technique condenses high-dimensional MFI data into two- or three-dimensional spaces, facilitating a more intuitive understanding of reactivity patterns. In the MDS plots, each point represents a specific HLA antigen, and the spatial distribution of these points reflects the degree of similarity or difference in their reactivity profiles.

A 2D MDS offers a simpler view with minimal loss of context. While 3D MDS offers a comprehensive view, 2D MDS simplifies data visualization without significant loss of contextual information, making it more accessible for clinical teams who do not specialize in data science. The 2D plots are especially useful for quick assessments and presentations, where clarity and immediacy are paramount.

In both 2D and 3D MDS (Figures 4 and 5, respectively), several consistent relationships are evident, for example, A*25:01, A*26:01, A*34:01, A*34:02, A*43:01, A*66:01, and A*66:02 separating from non-A10 serogroup A-locus specificities.

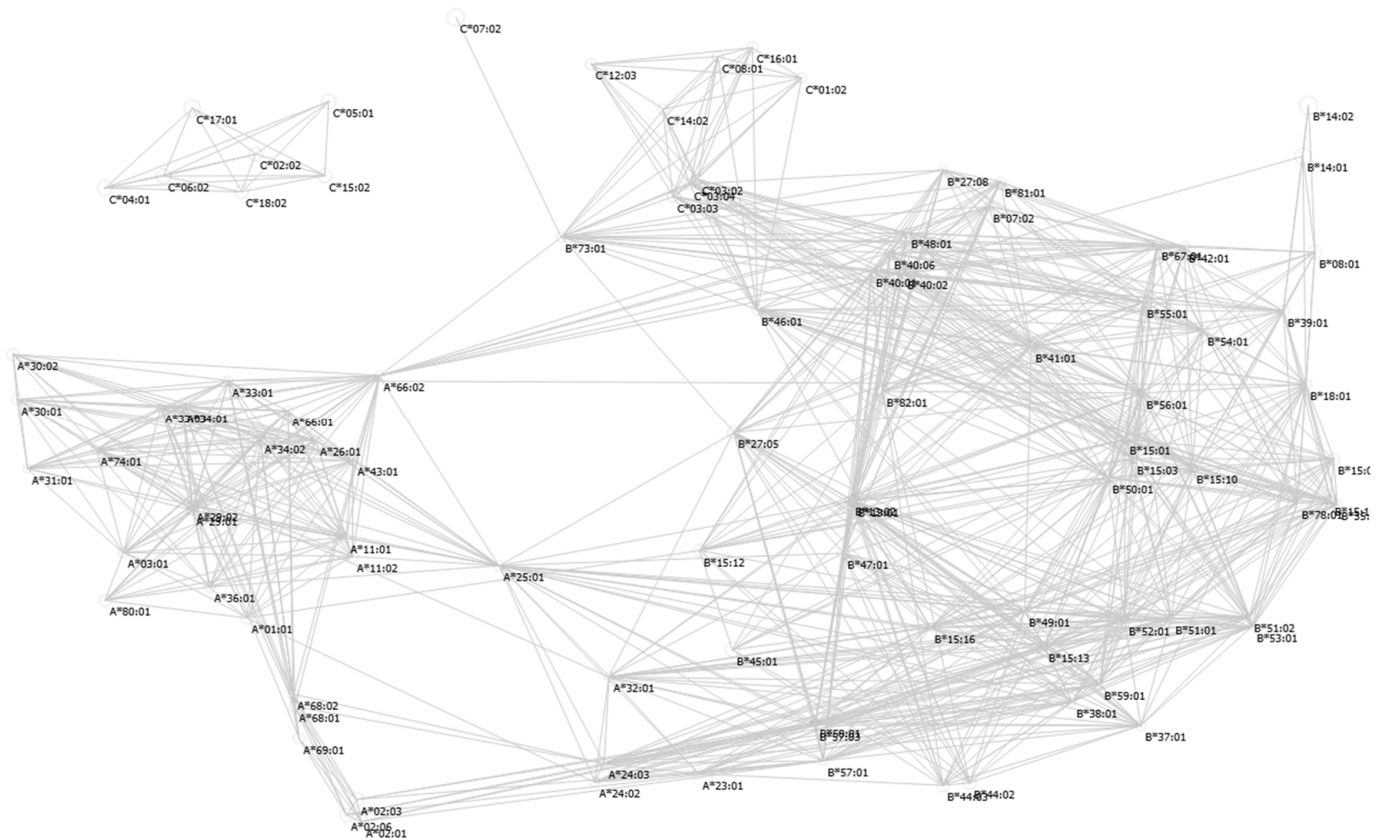


Figure 4. 2D multidimensional scaling (MDS) of anti-HLA Class I reactivity (SAB MFI). Each point is a Class I antigen (HLA-A*, -B*, -C*). Inter-point distances reflect dissimilarity in whole-profile MFI responses; nearer points share more similar reactivity and suggest potential cross-reactivity. The map provides an interpretable overview of antigen neighborhoods to support expert review and guide follow-up analyses.

These antibodies also originate from overlapping eplets (small amino acid configurations on molecules like HLA, recognized by antibodies) that are shared between them (e.g., 62RR, 62RNR, 62RN, 63NN, 62RRN, and 62RTN). In both cases, MDS was able to identify patterns related to either CREG or eplets. It is also interesting to note that antibodies directed to the HLA-C locus form a cluster (Figure 4). In fact, when taking into consideration

the eplets present in some of these alleles, a large majority of them are only present in the HLA-C locus (e.g., 1C, 69 RA, 73AN, 80K, 219W, among others).

3D MDS Visualization with Multiple Connections

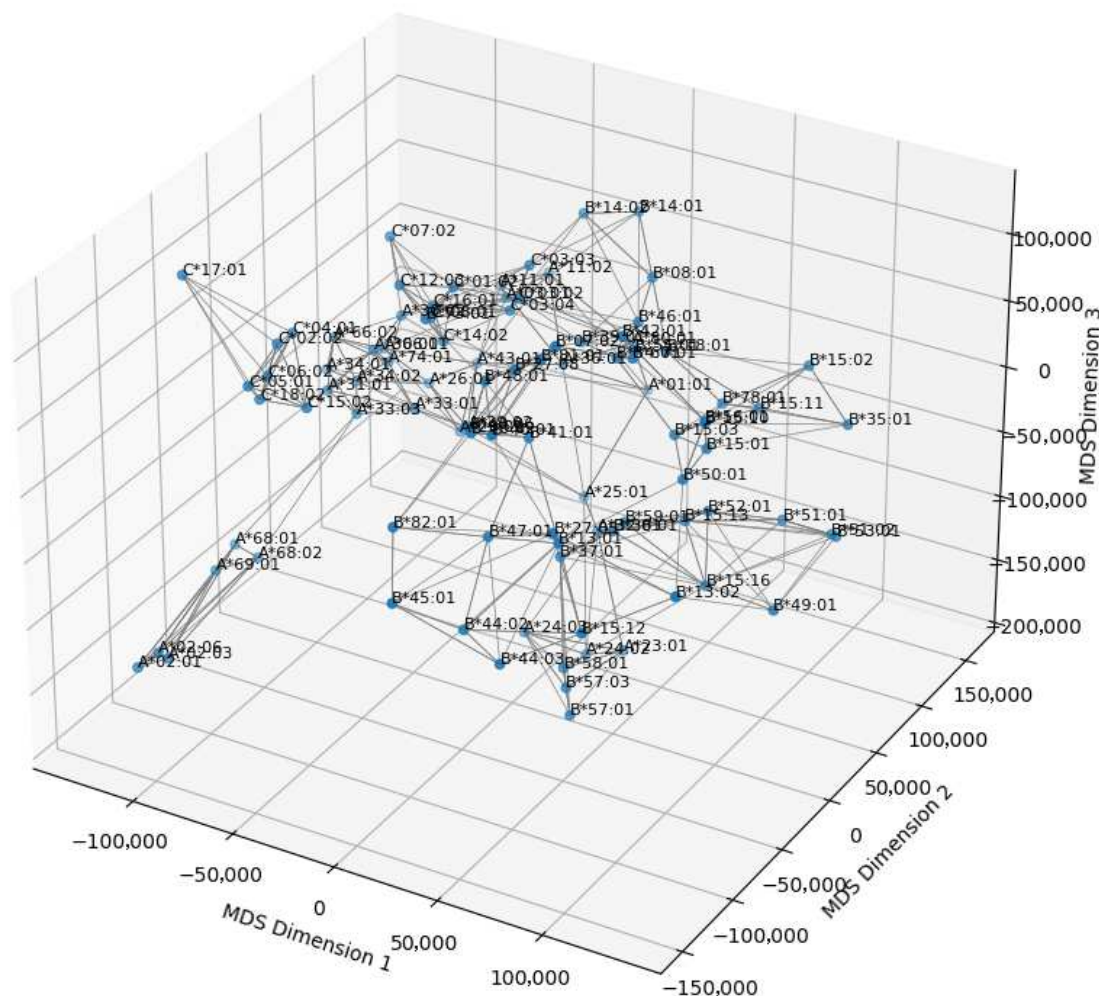


Figure 5. A 3D multidimensional scaling (MDS) of anti-HLA Class I reactivity (SAB MFI). Each point represents an HLA Class I antigen. Euclidean distances in the 3D embedding approximate dissimilarities in full-profile MFI responses; closer points indicate more similar reactivity and potential cross-reactivity. The third dimension helps separate partially overlapping neighborhoods seen in 2D, offering a clearer view of mesoscale antigen structure to support expert review and follow-up analyses.

3.4. Correlation Analysis

The correlation analysis explored the relationships among anti-HLA antibodies by assessing the strength and direction of their associations. The analysis involved calculating correlation coefficients, where positive correlations suggest that, as one antibody's reactivity increases, so does that of another antibody. This pattern indicates a potential similarity in reactivity, indicating cross-reactivity or similar binding characteristics. However, negative correlations suggest an inverse relationship, indicating that increased reactivity for one antibody is associated with decreased reactivity for another. Studying these correlations allows for the identification of groups of antibodies with common reactivity profiles, providing insights into their collective behavior.

In this study, two groups of correlations were evaluated. Thresholded views (≥ 0.95 and 0.90 to 0.95) are presented for readability; a complete set of pairwise correlation coefficients is available upon reasonable request. R^2 values equal to or greater than 0.95 were observed in Figure 6 (e.g., A*02:03 vs. A*02:06, A*29:01 vs. A*29:02, B*40:02 vs. B*40:06, and B*40:01 vs. B*40:06).

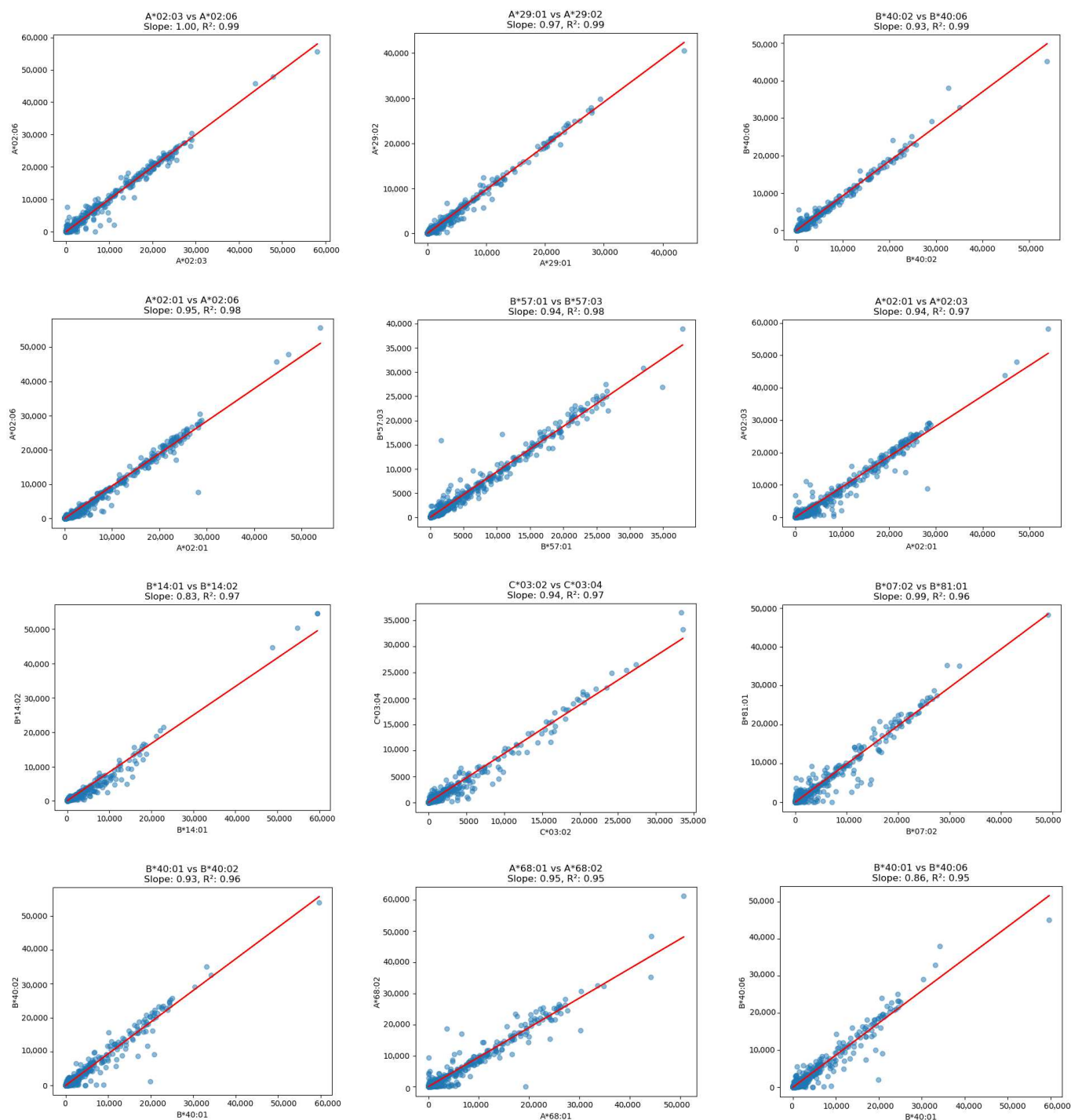


Figure 6. The correlation plot depicts the relationships between various anti-HLA antibodies. Each point represents a unique antibody MFI. A positive correlation suggests that two antibodies tend to show similar reactivity patterns, indicating potential cross-reactivity or similar binding characteristics, only correlations higher than or equal to 0.95.

These correlations involved antibodies directed against antigens in the same antigenic group as well as antibodies from different but closely related antigen groups (e.g., B*07:02 vs. B*81:01). These strong correlations suggest a high degree of similarity in the reactivity profiles of these antibodies, indicating that they likely bind to similar or shared epitopes. Identifying these correlations is crucial for understanding cross-reactive antibody responses.

Additionally, a second group of correlations with R^2 values between 0.90 and 0.95 included antibodies directed to serotypes rather than specific alleles (e.g., B*51:02 vs. B*53:01, B*57:01 vs. B*58:01). Although slightly weaker, these correlations still provide valuable insights into the patterns of antibody reactivity and potential cross-reactivity. Figure 7 shows that reactivity toward allelic groups is less evident within this range, with several correlations observed in serotypes rather than alleles.

Notably, a high correlation between antibodies directed to C*03:03 and C*03:04, which are characterized by permissible HLA mismatches in hematopoietic stem cell transplantation [28].

To avoid a one-sided view that only emphasizes sensitivity (i.e., identifying high correlations among presumed same-group antigens), also examined the complementary perspective. Specifically, among antigen pairs known to share CREG membership or high-confidence eplet overlap, the analysis quantified the fraction displaying very high correlation ($r \geq 0.95$) or moderately high correlation (0.90–0.95). Conversely, among antigen pairs not sharing these priors, the analysis measured how often they nonetheless reached high correlation values. These proportions, reported in Supplementary Table S2, contextualize correlation-based inference. The results confirmed enrichment of high- r pairs within known immunogenetic groups, while false-positive correlations among non-overlapping pairs were rare but not absent.

As detailed in Section 2.8, a complementary unsupervised model to test robustness was applied; the following subsections report the selected solution, its stability, network communities, cross-method consensus, and visual embeddings.

3.5. Advanced Machine-Learning Analyses and Robustness Checks

To assess whether the cross-reactivity structure reported above is robust to modeling choices and to expand beyond conventional MDS/HCA, the analysis implemented a complementary ML pipeline comprising (i) alternative distance formulations, (ii) multiple clustering families with principled model selection, (iii) stability analysis via bootstrap resampling, (iv) graph-based community detection on a sparsified correlation network, (v) non-linear embeddings for visualization, and (vi) an unsupervised consensus across methods. All analyses were performed on the antigen-by-test matrix after z-scaling features (antigen profiles) to mitigate scale effects and to partially control for assay-to-assay variability.

3.5.1. Distance Robustness and Dimensionality Reduction

Because Euclidean distance can overweight absolute intensity differences in semi-quantitative MFI data, a correlation-based dissimilarity (1–Pearson r) alongside the Euclidean geometry used for MDS/HCA was evaluated. PCA on z-scored profiles captured the dominant variability in a low-dimensional space (up to 20 PCs; Figure 8a; Supplementary Table S3. Variance explained by principal components), and all downstream clustering methods operated on this representation to standardize inputs across models.

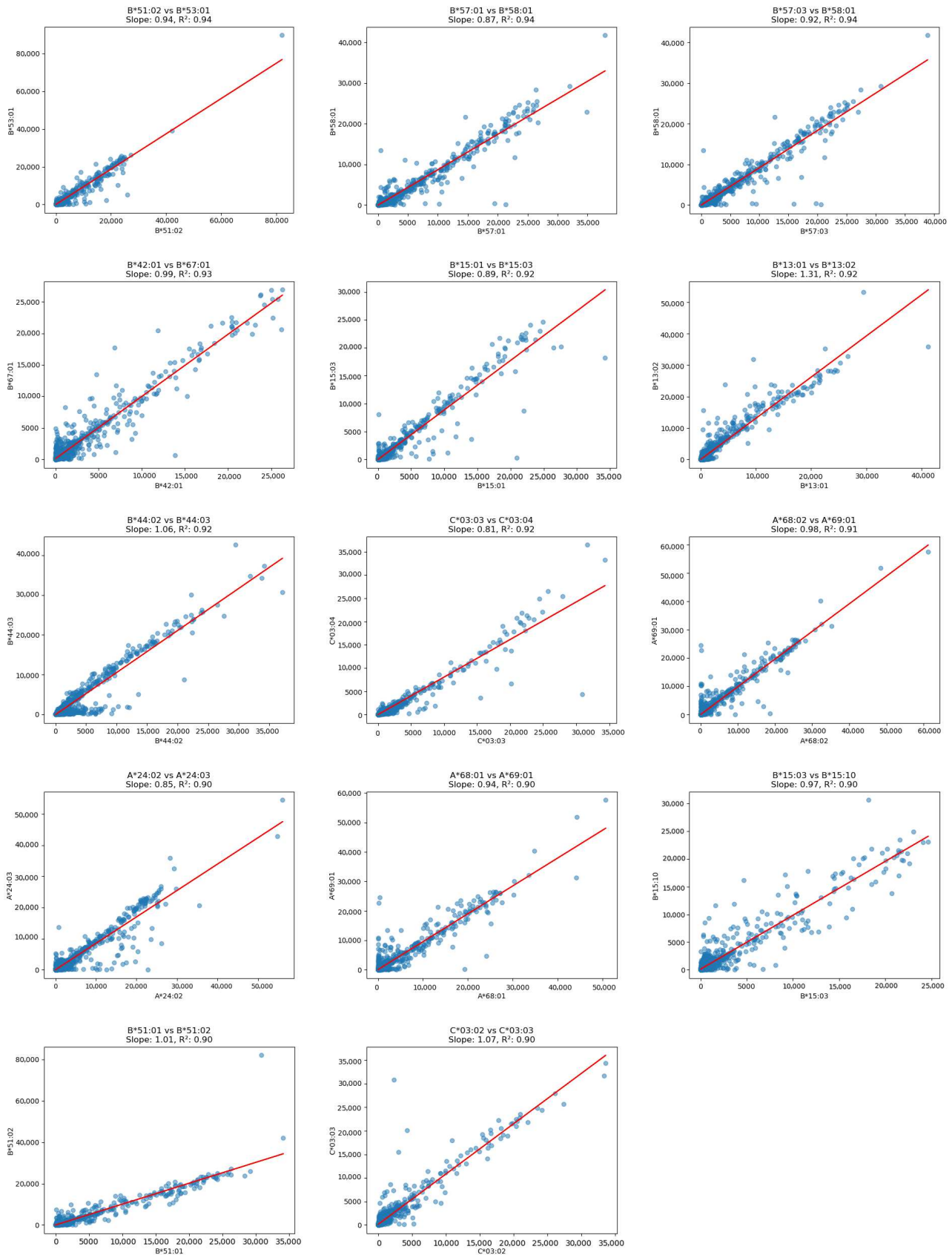


Figure 7. The correlation plot depicts the relationships between various anti-HLA antibodies. Each point represents a unique antibody MFI. A positive correlation suggests that two antibodies tend to show similar reactivity patterns, indicating potential cross-reactivity or similar binding characteristics, with correlations higher than or equal to 0.90 and lower than 0.95.

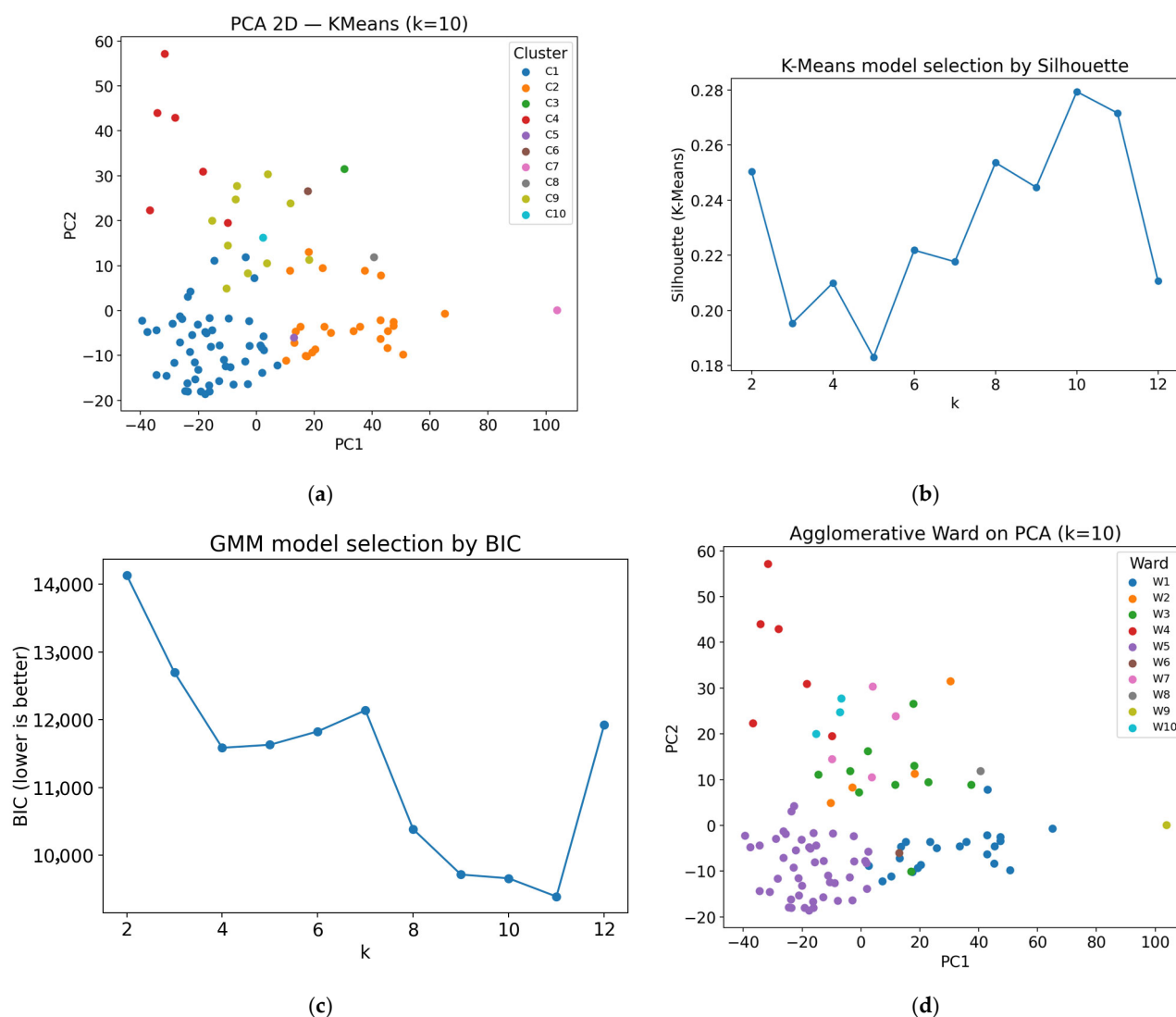


Figure 8. Cluster selection and low-dimensional geometry of antigen reactivity. (a) PCA (PC1–PC2) of z-scored SAB MFI profiles, colored by the selected “winner” clusters ($k = 10$; K-Means); (b) K-Means model selection: mean Silhouette vs. k (peak supports chosen k); (c) GMM model selection: BIC vs. k (minimum indicates preferred complexity); (d) Robustness check: Agglomerative (Ward) labels shown in the same PCA space.

3.5.2. Unsupervised Model Selection and Cluster Estimation

This study fit partitioning (K-Means), model-based GMMs, spectral, and agglomerative algorithms. For K-Means, the number of clusters (k) was chosen by maximizing the mean Silhouette on the PCA space; for GMMs, the analysis minimized the Bayesian Information Criterion (BIC). Across $k = 2$ – 12 , both criteria pointed to comparable solutions, yielding a “winner” model selected by the better Silhouette at the agreed k Figure 8b,c. The corresponding 2-D PCA visualization is shown in Figure 8d.

3.5.3. Clustering Stability

To assess robustness to sampling variability, five bootstrap resamples of 80% of tests (with replacement) were performed. For each resample, the selected model was refit twice with different seeds, and adjusted mutual information (AMI) between labelings was computed. AMI values ranged from 0 to 1, with higher values reflecting greater reproducibility. The median AMI was approximately 0.70, range 0.55–0.74 (Supplementary Table S4), indicating that cluster assignments are stable beyond initialization and sampling noise.

3.5.4. Alternative Clustering Families

This study compared (i) Agglomerative/ward on Euclidean PCA space (Figure 9a), (ii) Agglomerative/average on correlation distance $1-r$ (Figure 9b), and (iii) Spectral clustering on a correlation-derived affinity (Figure 9c). Performance metrics (Silhouette, Calinski–Harabasz, Davies–Bouldin) were computed consistently across methods; results are summarized in Supplementary Table S5. Comparative performance of clustering methods. These complementary approaches reproduce similar mesoscale groupings, supporting robustness to distance/model choice. Higher Silhouette and Calinski–Harabasz scores indicate more compact and well-separated clusters, whereas lower Davies–Bouldin indicate better separation. The Winner_KMeans ($k = 10$) and Consensus ($k = 10$) solutions achieve the strongest overall internal validity among tested approaches; HDBSCAN identifies only two dense groups and shows weaker separation, consistent with its conservative, density-based criterion.

3.5.5. Graph-Based Community Detection

In this study, an antigen–antigen similarity network from the standardized MFI profiles by retaining only positive Pearson correlations was constructed. To capture local neighborhoods while preserving global connectivity, the analyses formed a k -nearest-neighbor graph on the positive edges and merged it with a minimum-spanning-tree backbone. The final undirected, weighted graph (non-negative weights) was analyzed with modularity maximization. Greedy modularity returned coherent communities with high separation (modularity $Q \approx 0.61$), Figure 10, and these modules remained stable under alternative community algorithms. A spectral embedding of the graph is shown for visualization only; the axes have no direct biological meaning, while node colors indicate community membership. As a sensitivity analysis, the analysis repeated community detection with Louvain and Leiden Figure 11a,b ($Q \approx 0.61$ and $Q \approx 0.62$, respectively); both yielded partitions closely aligned with the greedy solution and similar modularity values, supporting the existence of mesoscale structure in the antibody-reactivity network. Node-level assignments and community summaries are provided in Community assignments are provided in Supplementary Table S6. Higher internal weight fractions indicate denser, more cohesive modules. Notably, communities 1–4 together account for $\sim 82\%$ of all internal edge weight, consistent with the large modules seen in Figure 10.

3.5.6. Non-Linear Embeddings and Density-Based Clustering

To explore local antigen neighborhoods beyond linear projections, t-SNE and UMAP on the PCA-transformed features were applied (Figure 12a,b). These non-linear embeddings highlighted contiguous clusters consistent with the winner solution while revealing plausible sub-structures within broad groups. Coloring by cluster membership showed that nearby antigens in PCA/MDS space also formed compact neighborhoods in UMAP/t-SNE, supporting visual robustness. Additionally, density-based clustering with HDBSCAN on the UMAP embedding identified dense cores and a small fraction of “noise” points, consistent with gradients of relatedness rather than fully separable partitions Figure 12c. These embeddings are intended for visual analytics and hypothesis generation, not for primary quantitative claims.

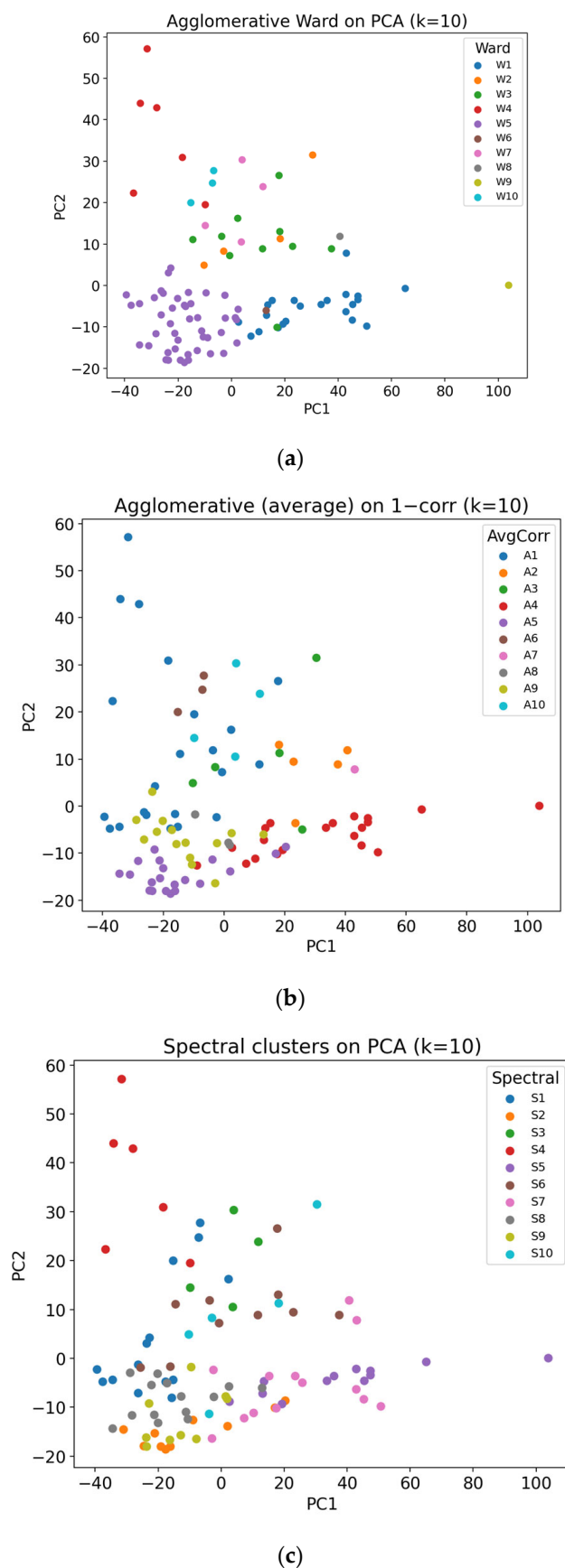


Figure 9. Alternative clustering families (composite). (a) Agglomerative Ward on Euclidean PCA space; (b) Agglomerative average on correlation distance ($1-r$); (c) Spectral clustering on a correlation-derived affinity. Points are antigens in the PC1–PC2 plane; colors indicate cluster membership ($k = 10$).

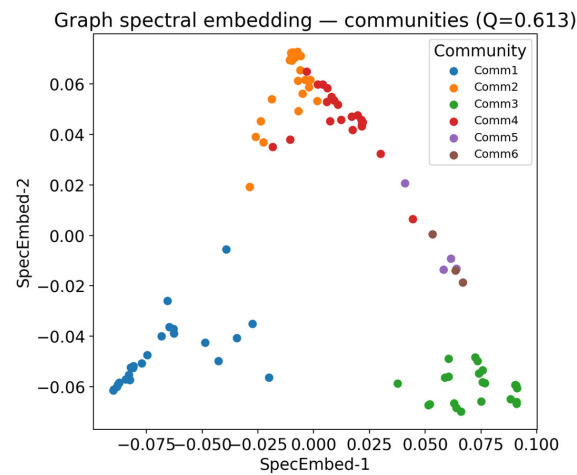


Figure 10. Graph-based communities in the antigen–antigen correlation network. Antigens (nodes) are connected by positive-correlation edges; a k-nearest-neighbor graph was combined with a minimum-spanning-tree backbone to ensure connectivity. The spectral layout is shown for visualization; colors denote communities obtained by greedy modularity maximization.

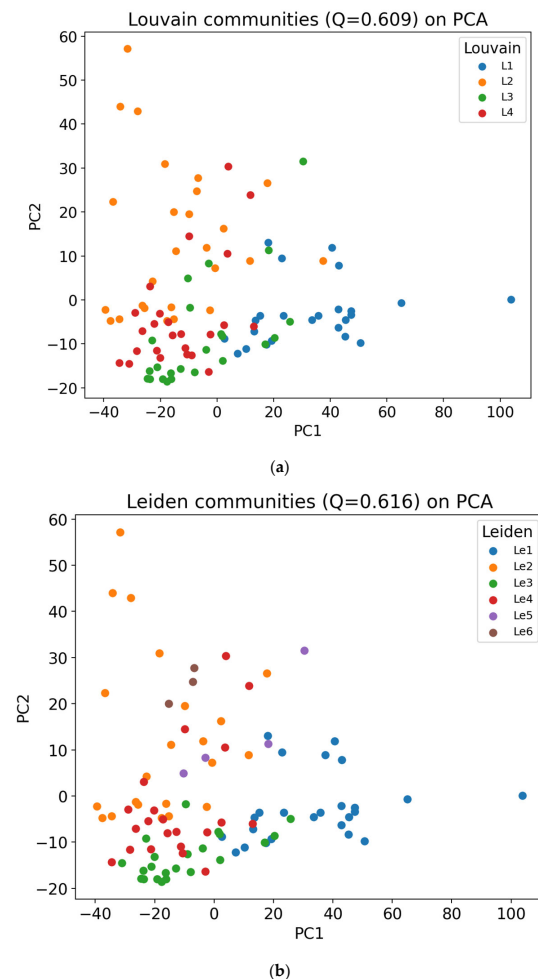
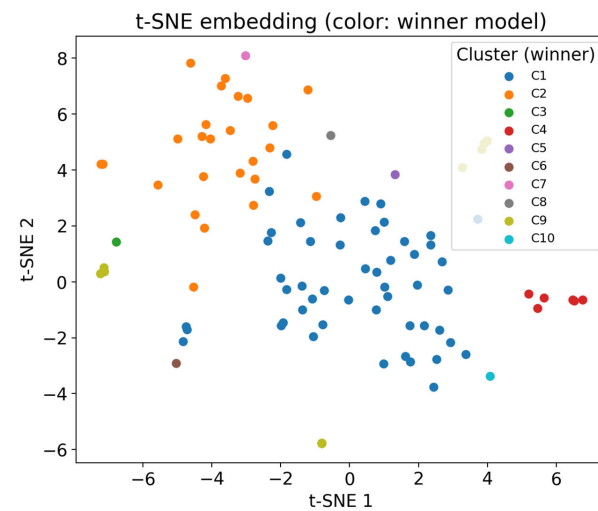
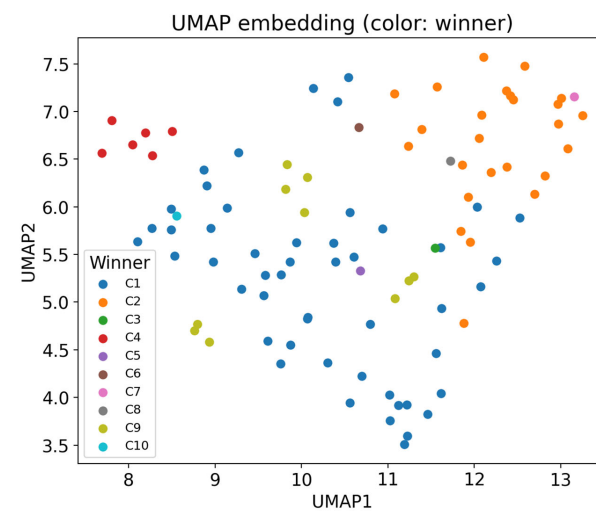


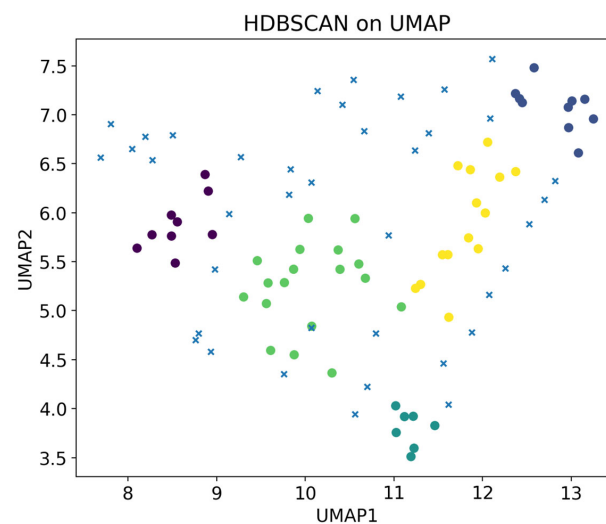
Figure 11. Graph-based communities in the antigen–antigen correlation network. (a) Projection of antigens onto PCA space (PC1–PC2), colored by Louvain communities (modularity $Q = 0.609$), illustrating concordance between graph-based modules and PCA structure, (b) Spectral embedding of the positive-correlation graph built by merging a k-nearest-neighbor graph with a minimum-spanning-tree backbone. Colors indicate communities identified by greedy modularity maximization (modularity $Q = 0.613$).



(a)



(b)



(c)

Figure 12. Non-linear embeddings and density-based clustering of antigen reactivity profiles. (a) t-SNE embedding of PCA-transformed antigen profiles, preserving local neighborhoods, (b) UMAP embedding highlighting both global and local antigen relationships, (c) HDBSCAN applied to the UMAP space, identifying dense antigen clusters while labeling sparse points as noise.

3.5.7. Consensus Clustering Across Methods

To assess the agreement between different clustering strategies, a consensus co-association matrix was constructed, which records the fraction of methods assigning each antigen pair to the same cluster. Hierarchical clustering of this matrix produced a consensus partition that closely resembled both the “winner” solution and graph-based communities. Importantly, the consensus matrix also highlights antigen pairs with inconsistent grouping across methods, providing a transparent measure of assignment stability. The numerical consensus matrix is provided as Supplementary Table S7.

3.5.8. Interpretation, Limitations, and Link to Prior Knowledge

Concordant groupings across Euclidean-, correlation-, and network-based formulations, together with non-trivial bootstrap AMI and modularity Q, suggest that the inferred structure is method-robust. Visual neighborhoods recapitulate known locus-level relationships (e.g., aggregation among HLA-C specificities and the proximity of A02-like and A68-like reactivities) while hinting at finer subdivisions that warrant prospective validation. Importantly, all modeling operated on z-scored MFI profiles to attenuate scale effects and mitigate batch variation; correlation-based distances and graph construction further down-weight absolute intensity shifts. Nevertheless, SAB MFI is semi-quantitative and can be influenced by reagent lots and assay conditions; therefore, our clusters should be interpreted as relative reactivity neighborhoods. External replication and linkage to clinical outcomes (e.g., crossmatch results, dnDSA emergence, rejection) will be essential to translate these groupings into decision support.

4. Discussion

This work proposes a transparent and auditable approach to mapping anti-HLA Class I antibody reactivity from SAB MFI profiles. Our primary goal was not to replace serological knowledge, but to structure it: to expose reproducible neighborhoods of reactivity that agree with established cross-reactive relationships while revealing finer modules that deserve prospective evaluation. To that end, this study intentionally favored explainable steps, distance-preserving embedding, and hierarchical clustering, and then stress-tested the same structure using complementary unsupervised families, including partitioning (K-Means), model-based clustering GMM, spectral clustering on correlation affinity, graph-based community detection, and density-based clustering on non-linear embeddings. This study additionally quantified robustness through bootstrap stability, AMI, and summarized cross-method agreement with a consensus co-association analysis. Together, these results indicate that the mesoscale organization of anti-HLA Class I reactivity in our dataset is not an artifact of a single metric or algorithm, but a recurrent signal that emerges across models, distances, and resampling. In addition, when matched against serologically defined CREGs and Bw4/Bw6 motifs, our data-driven groupings showed concordant patterns, underscoring that the computationally derived structure aligns with established immunochemistry [25,29,30].

4.1. What This Adds Beyond Standard MDS/HCA

MDS and HCA are standard tools; their value hinges on how consistently they recover meaningful modules and how well they integrate with biological priors. The present study extends a conventional MDS/HCA view in three ways.

First, this study triangulated the structure across method families. On a common low-dimensional representation (principal component space of z-scored antigen profiles), K-Means and GMM selections over $k \in [2,12]$ were guided by internal validity (Silhouette) and model evidence (BIC), with the “winner” chosen by Silhouette at the agreed k . Spectral

clustering, agglomerative Ward clustering (Euclidean), and agglomerative average linkage on correlation distance ($1-r$) returned comparable mesoscale partitions. While the labels differ in boundary cases, as expected because each algorithm optimizes a different criterion, the same neighborhoods repeatedly emerged, suggesting the presence of stable reactivity “regions” rather than method-specific artifacts.

Secondly, direct, code-level benchmarking against previously published pipelines [15,25,26], was not feasible for this dataset because (i) implementations and/or trained models are not publicly available end-to-end, (ii) several methods target different endpoints (e.g., epitope assignment or crossmatch prediction rather than unsupervised neighborhood discovery), and (iii) required inputs (patient-level covariates or laboratory calibrations) were not collected in our retrospective matrix. Instead, we triangulated performance using model-agnostic criteria on a common feature space (Silhouette/BIC), stability to resampling (bootstrap AMI), agreement across clustering families (consensus co-association), and a network modularity view. This design isolates the contribution of the data geometry from method-specific assumptions and, we believe, is the fairest comparison available in the absence of directly runnable reference code. We also mapped data-driven groups to established immunologic structure (CREGs/eplets) to provide biological face validity rather than purely numerical fit. (See Sections 3.5.2–3.5.7 and Supplementary Materials)

Thirdly, the organization as a network phenomenon was validated. Constructing an antigen–antigen similarity graph from positive correlations (k-nearest-neighbors with a minimum spanning tree backbone) and running community detection identified modularity-positive communities that overlapped with the partition found by the winner model and with the consensus co-association clustering. This cross-evidence (partitioning/model-based vs. spectral vs. graph communities) strengthens the claim that the observed modules reflect genuine neighborhood structure in the data.

Fourthly, stability was quantified. Bootstrap resampling of tests (80% with replacement) and re-fitting the selected model with independent seeds yielded non-trivial AMI, indicating that labels are not dominated by initialization or sample idiosyncrasies. This is important practically because clinical interpretation should rely on modules that persist under reasonable perturbations.

4.2. Relationship to CREGs, Eplet Sharing, and Prior Literature

Historically, CREGs and, more recently, eplet-level interpretations have provided mechanistic and serological bases for observed co-reactivity patterns. In our analysis, several neighborhoods align with these expectations: for example, recognizable A-locus and B-locus proximities and specific C-locus relationships; this analysis also found a high correlation of antibody pairs that bridge serological labels, where epitope exposure is plausibly shared. These findings are broadly consistent with previous heatmap-based examinations of SAB matrices and with studies linking antibody reactivity to shared motifs and epitope commonality. What this pipeline contributes is a unified and quantified view: neighborhood structure -that is visible in embedding, supported by multiple clustering families, summarized by consensus, and corroborated in a graph modularity framework. This integration moves the discussion from “visual plausibility” to “method-robust evidence”, still exploratory, but more reproducible and more readily comparable across cohorts [31,32].

4.3. Why Nine Groups? Statistical Selection and Biological Plausibility

These analyses emphasize that the number of groups was not fixed a priori. This study scanned $k = 2 \dots 12$ and combined complementary criteria: the internal validity of K-Means (Silhouette), the model evidence of GMM (BIC), and a final tie-break by Silhouette at the agreed k . The modal choice across these criteria defined the summary

scale used downstream. Importantly, when changed either (i) the distance family (Euclidean vs. correlation-based) or (ii) the clustering mechanism (Ward, average linkage, spectral), similar mesoscale organization was obtained. Graph communities fell near the same granularity, and the consensus co-association partition further confirmed cross-method agreement. Biologically, the resulting groups are consistent with known cross-reactivities and highlight modules that align with previous reports on epitope sharing and permissible mismatches [33,34].

4.4. Distance Metrics, Scaling, and Batch-Effect Concerns

SAB MFI is semi-quantitative and sensitive to technical factors (reagent lots, instrument settings, prozone, etc.). To reduce scale artifacts, each antigen's profile across tests (z-scoring) was standardized before modeling. Recognizing that Euclidean distance can overweight absolute intensity differences even after standardization, the analysis also examined correlation-based distances and affinities, which emphasize pattern similarity (co-reactivity shape) over level [19,35]. The principal neighborhoods persisted across these formulations, which reduces the risk that our conclusions merely reflect extremes in absolute MFI. Nonetheless, in this study, the authors interpret modules as relative reactivity neighborhoods rather than hard biological classes, precisely because residual batch effects cannot be ruled out retrospectively. Future work should include explicit lot/instrument covariates, replicate controls, and harmonization procedures to better partition technical from biological variation [35]. The present results should therefore be viewed as structure discovery that prioritizes robustness across models, distances, and resampling, not as final clinical categorization.

4.5. Graph Communities and Consensus: Why Two Views?

Clustering and community detection are related but not identical. Partitional/hierarchical methods assign every antigen to one of k groups, optimizing an explicit objective (e.g., variance decomposition). Community detection on a weighted graph instead seeks subgraphs with dense internal connectivity and sparser external ties. In our study, these views largely agreed: modularity-positive communities overlapped with the winner clustering and with the consensus partition derived from a co-association matrix (the fraction of methods that co-assign each pair). Where they differed, the disagreements were instructive, often marking boundary specificities or weakly connected nodes. In practice, it is recommended to use both: a crisp partition for reporting and an explicit network view to flag uncertain boundaries and “bridge” antigens. This is closer to the way clinicians reason about risk, continuous proximity, and neighborhoods than a single hard labeling [36].

4.6. Non-Linear Embeddings and Density-Based Clustering

This study used t-SNE and UMAP to visualize local neighborhoods in 2D, coloring by the winner labels. These methods are primarily for visualization; they are sensitive to hyperparameters and do not preserve global distances. Still, they help identify dense cores and outliers. As an exploratory complement, HDBSCAN on the UMAP space provided a density-based perspective that can label borderline points as noise; in our data, the coherent cores corresponded well to clusters seen in the PCA/MDS space. Consistency across these qualitatively different lenses increases our confidence that the neighborhoods are not an artifact of one projection or one optimization target [37].

4.7. Clinical Interpretation and Potential Utility

From a translational standpoint, neighborhood-aware maps of anti-HLA Class I reactivity can assist in several ways:

- i. Summarization of complexity. Highly sensitized patients often show broad reactivity. A neighborhood map can anchor the interpretation around a few modules rather than dozens of isolated specificities, making case conferences and multidisciplinary discussions more efficient.
- ii. Risk heuristics for unacceptable antigens. If a patient has strong reactivity in a given module, the neighborhood can highlight adjacent specificities that may pose risk, even if they are just below traditional MFI thresholds, guiding confirmatory testing (e.g., dilution series, EDTA treatment) or cautious listing decisions.
- iii. Hypothesis generation for outcomes research. Neighborhood membership, graph centrality, and consensus stability are testable features. In prospective datasets with linked clinical endpoints (crossmatch, de novo donor-specific antibody, rejection), one can evaluate whether these features add predictive value beyond absolute MFI thresholds and historical sensitization metrics.
- iv. Mechanistic insight. Modules that cut across serologic labels but share reactivity motifs suggest epitope-level commonalities. Closer integration with curated eplet sets and structural modeling would sharpen this link, potentially supporting epitope-aware matching policies in select scenarios.

This study stresses that these are hypotheses enabled by a robust exploratory structure; clinical deployment requires prospective validation.

4.8. Limitations

This is an exploratory, retrospective analysis of a single dataset without linked clinical outcomes. As such, this study cannot claim prognostic utility or clinical non-inferiority relative to existing risk models. Technical factors in SAB (lot/instrument effects, prozone) may still influence structure despite z-scoring and correlation-based formulations. Clustering methods—whether partitioning, model-based, or density-based—impose discrete labels on what is likely a continuum of relatedness; boundary antigens can shift with additional samples, and HDBSCAN’s “noise” labeling, while informative, complicates crisp reporting. The work is antigen-centric rather than epitope-resolved; closer integration with eplet annotations, structural modeling, and experimentally validated epitope maps would provide stronger mechanistic anchoring. Finally, although consensus clustering helps synthesize results across algorithms, it cannot fully eliminate method-specific artifacts; low co-association values mark unstable antigen pairs and should be interpreted cautiously [38–40]. Prospective validation in independent cohorts with linked clinical endpoints (e.g., rejection phenotypes, graft outcomes) will be needed to establish generalizability and translational value.

4.9. Future Directions

Three lines of work naturally follow. First, prospective validation in multi-center cohorts with harmonized SAB protocols and linked outcomes should test whether neighborhood-aware features (module membership, centrality, consensus strength) improve the prediction of crossmatch results, de novo donor-specific antibody, and rejection beyond absolute MFI thresholds and historical sensitization. Second, epitope-aware integration should align modules with curated eplet sets and structural motifs, quantifying where neighborhood boundaries map to epitope differences versus assay-level phenomena; disagreement cases will be especially informative. Third, decision-support prototyping can embed these maps into interactive dashboards that surface (i) a patient’s position in module space, (ii) nearby specificities likely to rise with immune pressure, and (iii) uncertainty indicators (e.g., low consensus or low graph cohesion) to guide confirmatory testing rather than categorical decisions.

5. Conclusions

This exploratory study delivers a transparent, reproducible workflow to map anti-HLA Class I reactivity from SAB MFI data. On a common PCA space of standardized antigen profiles, K-Means with $k = 10$ emerged as the winner by Silhouette and was convergent with Gaussian mixtures (BIC favored a comparable range of k), while spectral and agglomerative variants recovered similar mesoscale partitions. The winner solution was stable to resampling (bootstrap AMI ≈ 0.70), and both a consensus co-association summary and a graph-based community view (modularity Q) independently corroborated the same neighborhoods, indicating that the observed organization is not an artifact of a single distance metric or learner. Qualitatively, several data-driven groups aligned with established immunologic structure (e.g., CREGs/Bw4–Bw6 and shared eplet cues), providing face validity for the inferred cross-reactivity map.

Methodologically, the contribution is an auditable pipeline—rather than a new algorithm—that integrates model selection (Silhouette/BIC), robustness checks (bootstrap AMI, cross-family consensus), and a network perspective to turn a descriptive map into a method-robust summary suitable for comparison across cohorts. Practically, such a structure can help prioritize antigens for follow-up testing, focus expert review of unexpected cross-reactivity, and generate hypotheses for virtual crossmatch refinement.

This work remains hypothesis-generating. It analyzes a single retrospective matrix without linked clinical outcomes; technical variation in SAB may persist despite standardization and correlation-based formulations; and antigen-level clustering does not replace epitope-resolved analyses. Prospective validation on independent cohorts with harmonized laboratory controls and outcome data, and tighter integration with eplet annotations/structural models, is warranted. If confirmed, the approach may support consistent reporting of reactivity neighborhoods across centers and facilitate translational studies that connect antibody structure to clinical risk.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/aimed1010001/s1>.

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Abbreviations

The following abbreviations are used in this manuscript:

HLA	human leukocyte antigens
MFI	Mean Fluorescence Intensity
SAB	Single Antigen Bead
CREGs	Cross-Reactive Groups

DSAs	Donor-specific antibodies
MDS	Multidimensional scaling
HCA	Hierarchical cluster analysis
EDTA	Ethylenediaminetetraacetic acid
PCA	Principal component analysis
GMM	Gaussian Mixture Models
BIC	Bayesian Information Criterion
AMI	Adjusted mutual information
t-SNE	t-distributed stochastic neighbor embedding
UMAP	Uniform manifold approximation and projection
HDBSCAN	Hierarchical density-based spatial clustering of applications with noise

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